

The Portfolio Scenario:
National Healthcare Resource Allocation – Phase 2
(Data Analysis Project)

Data Analyst: Symone Thompson
Client/Sponsor: Public Health Infrastructure & Logistics Analytics
Target Asset: 02_ascending_low_volume.csv

Executive Summary & Objective:
This operational phase is designed to scrutinize and pinpoint distinctive geographic outliers in maternal health logs, prioritizing the absolute minimum non-zero volume parameters. By surfacing specific regional sectors with minimal birth distributions, the assessment establishes a strategic structural mapping of potentially underserved rural clinics or regions requiring enhanced data hygiene and infrastructure support.

Purpose:
This operational phase is designed to scrutinize and pinpoint distinctive geographic outliers in maternal health logs, prioritizing the absolute minimum non-zero volume parameters. By surfacing specific regional sectors with minimal birth distributions, the assessment establishes a strategic structural mapping of potentially underserved rural clinics or regions requiring enhanced data hygiene and infrastructure support.

Scope / Major Project Activities:
The major parts of this project. High-level steps, activities, or stages of the project, and a brief description for each.

Activity	Description
Environment Setup & Data Ingestion	Access the Google Cloud Platform sandbox, initialize a BigQuery workspace console, and connect to the public <code>sdoh_cdc_wonder_natality</code> data ecosystem. Ingestion

	of historical CDC Wonder Natality county logs via the Google Cloud Platform sandbox environment. Ingestion of historical CDC Wonder Natality county logs via the Google Cloud Platform sandbox environment.
Schema Assessment & Data Hygiene	Audit the technical structural layout of the <code>county_nativity</code> table definitions, verifying fields like <code>Year</code> , <code>County_of_Residence</code> , and <code>Births</code> integer metrics to ensure structural accuracy before running transformations. Execution of explicit sorting syntax utilizing standard ascending parameters on the birth metrics.
SQL Transformation & Aggregation	Develop and execute clean, well-commented SQL queries utilizing advanced sorting logic (<code>ORDER BY Births</code>) to isolate and rank the lowest-volume counties. Isolation of the top 10 absolute minimum non-zero regional entries for system validation.
Insight Generation & Business Reporting	Compare raw baseline data arrangements against sorted ascending output structures to deliver definitive geographic findings for strategic leadership decisions regarding rural outreach and clinical support.

This project does not include:

- **Total County Populations:** This analysis strictly measures raw birth volumes (`Births`) rather than calculating per capita birth rates against the total population of each county.
- **Gender Distinctions:** The scope isolates total infant delivery counts without filtering or segregating data based on the sex assigned at birth.
- **Stillborn and Mortality Metrics:** The dataset exclusively records live natality events; fetal deaths, stillbirths, and infant mortality tracking are completely excluded from this scope.
- High-density urban demographic sorting (reserved strictly for Phase 3 optimization tracks).
- Cross-sectional medical trend visualization or dashboard mockup deployment during this data-capture sprint.

Deliverables:

A specific list of things that this project will deliver.

Column Name	Data Type	Analytical Purpose
Year	DATE	Defines the specific temporal frame for tracking low-density maternal health records.
County_of_Residence	STRING	Pinpoints the precise geographic sector manifesting minimal natality data logs.
County_of_Residence_FIPS	STRING	Supplies the normalized regional code necessary for GIS visualization and mapping.
Births	INTEGER	The primary metric utilized to sequence the dataset by ascending volume thresholds.

Schedule Overview / Major Milestones:

Project execution commences on June 18th, 2026, structured across highly focused technical milestones.

Milestone	Expected Completion Date	Description/Details
Milestone 1:	June 18th, 2026	Query Drafting & Ascending Constraint Validation – Completed within 24 Hours.
Milestone 2:	June 19th, 2026	System Extraction to Google Sheets & CSV Formatting – Completed within 48 Hours.
Milestone 3:	June 21st, 2026	Deployment of Dataset 02 to Local Project Directories – Completed within 72 Hours.

*Estimated date for completion:

Final delivery, asset deployment, and repository verification are scheduled for **June 22nd, 2026**.

The Analytical Imperative:

In healthcare logistics, data is directly tied to human outcomes. This analysis transcends simple numbers on a screen; by uncovering a clear, ascending baseline in low-volume sectors like rural and remote counties, it provides decision-makers with the proactive runway required to shift medical funding, optimize clinical support networks, and ensure that maternal care resources are precisely deployed before local medical infrastructure faces critical bottlenecks. True data maturity turns historical logs into protective, life-saving strategies.